

DSA / Algorithms (Abdul Bari) - Day 27

45-day plan mapped to the Algorithms playlist

0/1 Knapsack (DP)

0/1 knapsack by DP - two methods and program.

Watch these Abdul Bari videos

- > 4.5 0/1 Knapsack - Two Methods - Dynamic Programming
- > 4.5.1 0/1 Knapsack Problem (Program) - Dynamic Programming

Key concepts

- Items are indivisible (take or leave).
- DP table over items x capacity.
- Set / tabular methods.

Key idea

```
dp[i][w]=max(dp[i-1][w], val[i]+dp[i-1][w-wt[i]]).
```

Tasks (1.5h)

- [] Watch both videos
- [] Fill the knapsack table
- [] Code the DP

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0